

Nima chaharbaghi

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Education

Master's degree, IEMDP Wireless telecommunication and IOT systems <i>École d'ingénieurs du numérique (ISEP)</i>  double degree student	2025 – 2026/09 PARIS, FRANCE
Master of Science in Telecommunications Engineering <i>UNIVERSITY OF BOLOGNA (UNIBO)</i>  CGPA:28.6/30	2024 – 2026/09 BOLOGNA, ITALY
Bachelor of Science in Electrical Engineering <i>ISFAHAN UNIVERSITY OF TECHNOLOGY (IUT)</i> 	2019 – 2023 ESFAHAN, IRAN

Interests

- 5G Prototyping
- Information Theory
- Software defined networking
- MIMO
- 5G Core Network
- ML for Wireless Communication
- Virtualization
- Localization

RESEARCH EXPERIENCE

Master's Thesis Researcher – Digital Twin & 5G Localization Under the supervision of Professor Florian Kaltenberger, I started my Master's thesis on the design and implementation of a digital twin infrastructure for the EURECOM GEO-5G testbed. • Performing GPU-accelerated ray-tracing simulations with NVIDIA Sionna RT to generate realistic channel impulse responses. • Integrating ray-traced channels into the OpenAirInterface (OAI) stack via the OAI Ray-Tracing Channel Emulator and rfsimulator. • Conducting uplink TDoA-based UE localization experiments using the OAI LMF module and evaluating localization accuracy.	2025/11 – 2026/09 SOPHIA ANTIPOLIS, FRANCE
Research Assistant, WiLab – University of Bologna Under the supervision of Professor Roberto Verdone and Professor Gianni Pasolini and in collaboration with Northeastern University, I contributed to the deployment and testing of a 5G system. Key accomplishments include: • Configured and tested the Radio Access Network (RAN) using the OpenAirInterface framework. • Worked within an OpenShift-based environment to manage system components and ensure stable operations. • Operated various USRPs (X410, B210) using the UHD library for hardware configuration, signal transmission, and device management. • Supported integration and troubleshooting efforts to improve network reliability and performance.	2025/01 – 2025/09 BOLOGNA, ITALY
Isfahan University of Technology IT Center Internship Experience: Network Manager • Developed proficiency in Linux administration at the LPIC-1 level. • Strengthened networking and troubleshooting abilities at the CCNA level. • Worked with OpenStack in cloud and virtualization projects, leading to better system performance and resource use.	2022/07 – 2022/09 ESFAHAN, IRAN

Projects

AI-Based Resource Prediction for 5G Core Networks (Open5GS + Kubernetes) 	2025
• Deployed a full 5G Core network using Open5GS and UERANSIM on Kubernetes. • Collected AMF CPU and memory metrics using Prometheus. • Applied Chronos-T5 (Tiny) for time-series forecasting and compared results with TimeGPT.	
Secure Video Transmission System Using H.264 and AES Encryption	2024
• Designed a system to receive and decode transport stream data transmitted in H.264 format. • Applied Advanced Encryption Standard (AES) to secure video streams, preventing unauthorized access. • Implemented decryption for both AES and H.264 encoded data to ensure smooth video playback. • Analyzed data flow, encryption, and decryption processes to optimize security and performance.	
Real-Time Hand Gesture Recognition Using mmWave Radar (Final Year Project)	2023
• Designed and implemented signal processing algorithms to extract gesture features. • Utilized machine learning techniques to classify and interpret hand gestures accurately. • Integrated the system with user interfaces for practical applications.	
Implementing Effects of AWGN on Constellation, Constant Complex Channel Gain, and Multipath Channel using MATLAB	2023
• Utilizing Gray code for constellation point mapping to enhance error resilience. • Generating random binary sequences and mapping them to complex constellations. • Evaluating signal power and noise power for different Signal-to-Noise Ratios (SNR). • Applying complex Gaussian noise to simulate real-world channel conditions. • Investigating the impact of constant complex channel gain on signal constellations.	

Languages

ENGLISH IELTS 6.5	PERSIAN Native	FRENCH A1
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Skills

Matlab	Python
C++	LaTeX
XILINX Vivado	OpenCV
Open Air Interface	CCNA
SRSran	Linux LPIC

References

Davide Dardari  , *Full Professor*
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Roberto Verdone  , *Full Professor*
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Gianni Pasolini  , *Associate Professor*
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Florian Kaltenberger  , *Associate Professor*
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